

Complete nonet interactions and invariant CKM geometry

A bounded 70-coefficient action, stable physical CP and mass-preserving coupling response

PerfectPower research draft / 6 October 2026 (UTC)

Exact operator counts and target equations, numerical isolated vacua and finite-deformation checks. The full allowed action retains four adjustable mixing directions.

Result and declared theory

A complete renormalizable scalar basis, an isolated CP-breaking vacuum and a fixed-spectrum response calculation now belong to one specified minimal theory. Exact character arithmetic gives 60 independent real scalar coefficients before the Higgs, of which 28 are mixed quartics. The neutral Higgs adds two self coefficients and eight portals, giving 70 coefficients and 21 physical neutral real scalar coordinates. Numerical minimization produces nondegenerate source spectra and physical weak CP without supplying a nominated angle or CKM magnitude. The Hermitian fermion action retains its termwise first-order one-loop mass-phase cancellation for the calculated scalar eigenmodes.

The complete allowed coupling space does not predict the golden coefficient or the 66-degree standard phase. Four locally independent mixing directions survive even when all six light quark masses are held fixed. This is a constructive local statement on the delivered CP-breaking branch, not a classification of every branch. The repository's polynomial engine supplies an exact, rephasing-invariant test of the desired observable locus and derives a conditional elliptic slice of it. The interaction and the target locus are calculated separately so that algebraic target structure cannot be mistaken for dynamical prediction.

The fields are new minimal real Hermitian nonets, not the former complex source multiplets or the fitted degree-18 locking theory. Each sector contains η , the nonet trace coordinate s and a traceless Hermitian matrix A . Both singlets and all eight real components of A are odd under that sector's Z_2 ; the other sector's Z_2 acts trivially. Both sectors share the diagonal family group $3.A_6$ and its previously certified generalized CP. Ordinary quarks are neutral triplets; vectorlike triplets carry their sector's odd charge. The Higgs is a family singlet and even under both charges. All kinetic metrics in this action are canonical bare metrics.

Complete scalar census

Write A as a sum of eight orthonormal Hermitian basis matrices with real coordinates a . Let N be a dot a . The four quadratic invariant functions in either sector are η squared, s squared, ηa and N . The quadratic adjoint covariants are ηa , $s a$ and the traceless part of A squared. The symmetric-square adjoint vector w has 36 orthonormal coordinates. Its finite-group projectors have ranks 1, 8, 8, 9 and 10 after imposing CP. The rank-ten channel combines the inequivalent five-dimensional channels exchanged by CP. The two eight-dimensional channels are inequivalent representations, rather than an arbitrary multiplicity-space choice.

Exact character averages over all 1,080 elements are evaluated in $Q(\sqrt{5}, \omega)$. Symmetric square of the real representation $8+1+1$ has four invariant singlets; its symmetric fourth power has twelve. The square of its symmetric-square character gives 29 mixed quartic singlets before CP. The difference of the two five-dimensional contractions is CP odd, leaving 28 even contractions. An independent numerical evaluation matrix for the entire displayed operator basis has rank 60. Numerical projector tensors are explicitly distinguished from the exact character dimensions.

The twelve self quartics per sector comprise the five degree-four monomials in η and s , the three degree-two singlet monomials times N , $\eta \text{Tr}(A^3)$, $s \text{Tr}(A^3)$, N^2 , and the self contraction in the extra finite eight-dimensional projector. The finite nine-dimensional self contraction equals $9 N^2$ divided by 40, so it cannot replace the independent extra-eight contraction. That polynomial dependence is important when translating the representation decomposition into a scalar basis.

The mixed quartics comprise sixteen products of quadratic singlets, nine dot products of quadratic adjoint covariants and three additional cross-projector contractions in the extra eight, nine and CP-paired five channels. The singlet projector and ordinary SU(3) adjoint projector are already included in the first two sets. The singlet and adjoint multiplicities therefore matter: five adjoint-only channels do not constitute the complete nonet potential. Independent sector Z2 charges forbid mixed quadratic terms and all odd-degree scalar interactions. Eight Higgs portals multiply the four quadratic singlets in each sector by H-dagger H.

A bounded vacuum without flavor targets

The executable search uses a declared seeded family of potentials. Six unit radial quartics dominate; the sum of absolute values of all remaining quartic coefficients is 0.12. Every displayed normalized quartic is bounded in absolute value by the fourth power of the total nonet-coordinate norm. Consequently the quartic part is bounded below by 7/150 times that norm to the fourth power. All sixty coefficients are nonzero. The Higgs quartic is 0.13, all eight portals are nonzero, and their two singlet forms and adjoint coefficients are positive. In canonical neutral coordinates the joint quartic lower bound is 0.01625 times the fourth power of the combined norm. These bounds establish boundedness and existence of a global minimum; they do not identify the selected local branch as that global minimum.

The scan stops at the first locally positive, nondegenerate branch with absolute J above 10 to the minus five. This criterion requests spontaneous physical CP breaking, without comparing any result to a flavor target. The selected seed is 20261010, start zero. All seventeen minimizations, complete coefficients and canonical coordinates are retained in the receipt. A root refinement solves all stationary equations after adding Higgs portals. The neutral scalar Hessian is positive, including its shallow angular direction: its smallest eigenvalue is 1.1262305e-05, the stationary gradient norm is below 10 to the minus twelve, and $J=0.0002622534621$. The exact generalized-CP transform gives an equal-energy partner with opposite J and the same masses and mixing magnitudes. CP invariance alone selects neither sign.

All quoted energies, coordinates, Hessian eigenvalues and vacuum responses are numerical. Exact character arithmetic, exact polynomial identities and the global analytic quartic bound have different guarantees. The retained search history is a reproducible existence calculation, not an exhaustive vacuum census.

Physical quark matching and mass-phase protection

Each sector has five real renormalizable fermion coefficients. Its Hermitian heavy-to-light source is $C=g\eta I+cs\ I+ca\ A$. With $a=v\ y$ and positive vectorlike mass m , the six-state Dirac matrix has blocks $a\ I$, zero, C and $m\ I$. Its determinant is $(a\ m)^3$, real and positive. CP sets the bare QCD theta parameter to zero; the spontaneous weak phase does not change this tree determinant.

In the eigenframe of C each family is a real two-state singular-value problem. This calculates the three light and three heavy masses, their ordering and the doublet weights without replacing the finite theory by its asymptotic limit. The light charged-current matrix is $\text{diag}(qu)\ U\ \text{diag}(qd)$, where U is the unitary source-frame overlap. Dividing by the calculated weights recovers U . The invariant CKM polynomial identities below apply to this unitary frame; they must not be applied directly to a nonunitary finite light current. The illustrative vacuum has six nonzero, nondegenerate light masses and nonzero J; it does not reproduce the measured hierarchy or CKM magnitudes.

For every neutral real scalar direction the variation of C is Hermitian. Diagonalizing the complete 21-coordinate scalar Hessian gives real orthogonal mixtures of both nonets, both eta fields and the Higgs. The paired chirality-flip products remain real for each eigenmode, giving zero imaginary first-order one-loop mass trace term by term. The numerical reality error at this branch is below 10 to the minus twelve. This verifies the existing algebraic theorem in the complete minimal scalar action. It does not establish two-loop protection, eliminate higher-dimensional fermion operators or derive the measured flavor parameters.

Fixed-mass observable freedom

The implicit vacuum derivative for a scalar coefficient is minus the inverse Hessian times that operator's gradient. All 28 mixed coefficients are tested. To hold six signed source eigenvalues fixed, six independent allowed quadratic coefficients compensate the mixed perturbation. The six-by-eight quadratic compensator has full row rank. Holding the signed eigenvalues and Higgs expectation value fixed preserves all six light masses, all six heavy masses and the current weights in the exact finite block formula.

The four observables are the squared magnitudes of U-us, U-cb, U-ub and U-td. The fixed-spectrum response has four nonzero singular values: 133.153996, 20.7912106, 1.14673727, 0.00111955388. Four selected independent mixed directions are then varied in both signs and all stationary and eigenvalue constraints are solved again. These finite solutions preserve the complete masses to numerical precision, retain positive scalar curvature and agree with the implicit response. Their J values and golden residuals change. The maximum complete-mass change is below 10 to the minus fourteen. The independently computed four-column finite response has smallest singular value 0.00057542912, exceeding its finite/implicit spectral discrepancy 0.00011204845. The smallest response direction is much weaker than the others, so finite-difference and Hessian accuracy are retained in the validation rather than inferred from a rank threshold alone.

This calculation identifies actual permitted interactions that move physical flavor at fixed mass. It is stronger than varying an unconstrained source matrix or changing a phase convention. The present symmetries provide a complete action with spontaneous CP, but they leave the golden and 66-degree relations adjustable. Additional restrictions would have to constrain this displayed coupling space; merely reproducing a target stationary root would not remove these directions.

Observable polynomial constraints

Set $a=|U-us|^2$, $b=|U-cb|^2$, $c=|U-ub|^2$, $d=|U-td|^2$ and $t=1-c$. Define $K=abc(t-a)(t-b)$ and $N=ab+c(t-a)(t-b)-d$ t squared. Then the exact unitary identity is

$$4t^2j^2 + N^2 - 4K = 0.$$

The standard CKM phase satisfies $\cos(\delta)=N$ divided by twice the square root of K. This extraction uses magnitudes and J, without assigning unphysical quark phases. The scalar vacuum angle, standard delta, unitarity-triangle gamma and strong theta are distinct quantities.

For the nominated depth $C=\sqrt{c}$ times t divided by square root of ab, put $X=C$ squared. Golden depth $C=\phi$ to the minus two obeys X squared minus seven X plus one equals zero. The observable necessary equation is

$$c^2t^4 - 7abct^2 + a^2b^2 = 0.$$

The small-root branch requires $0 < ct^2 < ab$. For $\delta = 66$ degrees, $x = 2 \cos(\delta)$ has minimal polynomial $x^8 - 7x^6 + 14x^4 - 8x^2 + 1 = 0$. Eliminating x gives

$$N^8 - 7N^6K + 14N^4K^2 - 8N^2K^3 + K^4 = 0.$$

Branch conditions are $K > 0$, $N > 0$, $J > 0$ and $16K/25 < N^2 < 1681K/2500$. Exact rational root isolation proves there is one phase root in the corresponding x interval $[4/5, 41/50]$. The receipt and tests reject the selected vacuum as a golden/66-degree prediction even though it satisfies the general unitary CP identity. Denominator-cleared equations without these branch conditions would admit unwanted roots.

The polynomial machinery and the elliptic slice

Fix c , X and a nontrivial standard phase independently. Write $x = a/t$ and $k = ct^2/X$. The remaining magnitude b is k/a , and the exact J relation becomes

$$(x/J)^2 = k c \sin^2 \delta x(1-x)(x-c/X).$$

For $\lambda = c/X$ different from zero and one, the cubic has three distinct roots. This is a quadratic twist of the Legendre genus-one curve with j -invariant $256(1-\lambda+\lambda^2)^3$ cubed divided by $\lambda^2(1-\lambda)^2$. For $0 < \lambda < 1$ the physical magnitude interval is $\lambda < x < 1$. The actual repository PolynomialCurve engine derives a two-dimensional de Rham connection from the cubic, and exact symbolic arithmetic checks the discriminant and coordinate substitution.

The fixed c slice is a conditional target locus, not a symmetry-derived selected vacuum. Its connection transports mathematical periods as λ changes; it does not select a quark frame or a gauge coupling. Rational slice examples are declared separately from the algebraic golden value. The machinery is useful for exact elimination, branch isolation and constraint geometry, while the allowed-interaction response supplies the physical prediction test.

There is also an exact restricted obstruction. At fixed nondegenerate spectra, four mixed traces $\text{Tr}(RT)$, $\text{Tr}(RT^2)$, $\text{Tr}(R^2T)$ and $\text{Tr}(R^2T^2)$ are independent magnitude coordinates, with Jacobian determinant $\Delta(R)^2 \Delta(T)^2$. Ordinary continuous-SU(3) trace quartics reduce to an affine function of those moments plus one squared moment. Their Hessian has rank at most one, so they cannot supply an isolated interior CP-breaking minimum on that four-dimensional slice. The finite-group quartics used above lie outside this restricted class. This distinguishes a necessary finite interaction from another unconstrained trace potential.

Reproduction and verification

The scientific receipt is `receipts/m22_interactions/valentiner_nonet_joint.json`. The scalar module is `python/perfectpower/nonet_potential.py`; the target geometry module is `python/perfectpower/ckm_constraint_geometry.py`. Run `PYTHONPATH=python OPENBLAS_NUM_THREADS=1 python python/develop_valentiner_nonet_joint.py` to rebuild it. Focused tests cover exact counts, operator independence, analytic gradients, group and CP covariance, boundedness, the Higgs vacuum, finite currents, scalar-mode mass-phase reality, fixed-spectrum response and exact target/curve identities. Numerical replay is checked from a fresh archive extraction, with its environment and tolerances recorded in the validation receipt.